Matrix Calculus for Least Squares and MAP

A short guide to vector gradients

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Why this matters

In least squares, the objective is one number:

$$J(\boldsymbol{w}) = \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}\|_2^2.$$

The unknown parameter is a vector:

$$\boldsymbol{w} = (w_1, \ldots, w_d)^\mathsf{T}.$$

Differentiating with respect to $\boldsymbol{w}$ means asking how J changes when each coordinate changes. Therefore,

$$\nabla_{\boldsymbol{w}} J = \begin{bmatrix} \partial J/\partial w_1 \\ \vdots \\ \partial J/\partial w_d \end{bmatrix}.$$

A scalar objective differentiated with respect to a vector gives a gradient vector.

Gradient shape

If

$$f : \mathbb{R}^d \to \mathbb{R},$$

then f takes a vector input and returns a scalar. In this course, we use the column-gradient convention:

$$\nabla_{\boldsymbol{w}} f = \begin{bmatrix} \partial f/\partial w_1 \\ \partial f/\partial w_2 \\ \vdots \\ \partial f/\partial w_d \end{bmatrix} \in \mathbb{R}^d.$$

The gradient has the same shape as $\boldsymbol{w}$:

$$\boldsymbol{w} \in \mathbb{R}^d, \qquad \nabla_{\boldsymbol{w}} f \in \mathbb{R}^d.$$

Common mistake

Some books use row-gradient notation. The formulas may look transposed, but the meaning is the same.

Derivative rules

Let $\boldsymbol{a}$ and $\boldsymbol{A}$ be constants with respect to $\boldsymbol{w}$.

Expression	Gradient with respect to $\boldsymbol{w}$
c	$\mathbf{0}$
$\boldsymbol{a}^\mathsf{T}\boldsymbol{w}$	$\boldsymbol{a}$
$\boldsymbol{w}^\mathsf{T}\boldsymbol{a}$	$\boldsymbol{a}$
$\boldsymbol{w}^\mathsf{T}\boldsymbol{w}$	$2\boldsymbol{w}$
$\boldsymbol{w}^\mathsf{T}\boldsymbol{A}\boldsymbol{w}$	$(\boldsymbol{A}+\boldsymbol{A}^\mathsf{T})\boldsymbol{w}$

If $\boldsymbol{A}$ is symmetric,

$$\nabla_{\boldsymbol{w}}(\boldsymbol{w}^\mathsf{T}\boldsymbol{A}\boldsymbol{w}) = 2\boldsymbol{A}\boldsymbol{w}.$$

Squared norm

Start with the simplest quadratic:

$$f(\boldsymbol{w}) = \boldsymbol{w}^{\mathsf{T}}\boldsymbol{w}.$$

Written coordinate by coordinate,

$$f(\boldsymbol{w}) = w_1^2 + \cdots + w_d^2.$$

The derivative with respect to one coordinate is

$$\frac{\partial f}{\partial w_k} = 2w_k.$$

Stacking all coordinates gives

$$\nabla_{\boldsymbol{w}} f = \begin{bmatrix} 2w_1 \\ \vdots \\ 2w_d \end{bmatrix} = 2\boldsymbol{w}.$$

The gradient is a vector because it collects one partial derivative for each component of w.

Quadratic form

Consider

$$f(\boldsymbol{w}) = \boldsymbol{w}^{\mathsf{T}} \boldsymbol{A} \boldsymbol{w}.$$

In coordinates,

$$f(\boldsymbol{w}) = \sum_i \sum_j w_i A_{ij} w_j.$$

When differentiating with respect to w_k, w_k can appear in two places:

$$w_i A_{ij} w_j.$$

It can be the left factor w_i, or the right factor w_j. This gives two contributions:

$$\frac{\partial f}{\partial w_k} = \sum_j A_{kj} w_j + \sum_i w_i A_{ik}.$$

Therefore,

$$\nabla_{\boldsymbol{w}} f = (\boldsymbol{A} + \boldsymbol{A}^{\mathsf{T}}) \boldsymbol{w}.$$

Symmetric case

If $\boldsymbol{A}$ is symmetric, then

$$\boldsymbol{A} = \boldsymbol{A}^{\mathsf{T}}.$$

The quadratic-form rule becomes

$$\nabla_{\boldsymbol{w}}(\boldsymbol{w}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{w}) = (\boldsymbol{A} + \boldsymbol{A})\boldsymbol{w} = 2\boldsymbol{A}\boldsymbol{w}.$$

This is why we often see a factor of 2. In least squares,

$$\boldsymbol{A} = \boldsymbol{X}^{\mathsf{T}}\boldsymbol{X}.$$

This matrix is symmetric because

$$(\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X})^{\mathsf{T}} = \boldsymbol{X}^{\mathsf{T}}\boldsymbol{X}.$$

The factor 2 appears because both copies of w contribute, and the two contributions are equal when A is symmetric.

Least squares setup

Stack all observations into a matrix and a vector:

$$\boldsymbol{X} \in \mathbb{R}^{n \times d}, \qquad \boldsymbol{y} \in \mathbb{R}^{n}, \qquad \boldsymbol{w} \in \mathbb{R}^{d}.$$

The prediction vector is

$$\hat{\boldsymbol{y}} = \boldsymbol{X}\boldsymbol{w} \in \mathbb{R}^{n}.$$

The residual vector is

$$\boldsymbol{r} = \boldsymbol{y} - \boldsymbol{X}\boldsymbol{w} \in \mathbb{R}^{n}.$$

Least squares minimizes the squared residual length:

$$J(\boldsymbol{w}) = \|\boldsymbol{r}\|_2^2 = \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}\|_2^2.$$

The objective is scalar, but the variable being optimized is the vector w.

Expand least squares

Start from

$$J(\boldsymbol{w}) = \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}\|_2^2.$$

By definition of squared norm,

$$J = (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w})^\mathsf{T}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}).$$

Expanding gives

$$J = \boldsymbol{y}^\mathsf{T}\boldsymbol{y} - \boldsymbol{y}^\mathsf{T}\boldsymbol{X}\boldsymbol{w} - \boldsymbol{w}^\mathsf{T}\boldsymbol{X}^\mathsf{T}\boldsymbol{y} + \boldsymbol{w}^\mathsf{T}\boldsymbol{X}^\mathsf{T}\boldsymbol{X}\boldsymbol{w}.$$

The two middle terms are both scalars and equal, so

$$J = \boldsymbol{y}^\mathsf{T}\boldsymbol{y} - 2\boldsymbol{w}^\mathsf{T}\boldsymbol{X}^\mathsf{T}\boldsymbol{y} + \boldsymbol{w}^\mathsf{T}\boldsymbol{X}^\mathsf{T}\boldsymbol{X}\boldsymbol{w}.$$

Differentiate least squares

The expanded objective is

$$J = \boldsymbol{y}^\mathsf{T}\boldsymbol{y} - 2\boldsymbol{w}^\mathsf{T}\boldsymbol{X}^\mathsf{T}\boldsymbol{y} + \boldsymbol{w}^\mathsf{T}\boldsymbol{X}^\mathsf{T}\boldsymbol{X}\boldsymbol{w}.$$

The first term is constant in $\boldsymbol{w}$:

$$\nabla_{\boldsymbol{w}}(\boldsymbol{y}^\mathsf{T}\boldsymbol{y}) = \mathbf{0}.$$

For the linear term, let

$$\boldsymbol{a} = \boldsymbol{X}^\mathsf{T}\boldsymbol{y}.$$

Then

$$\nabla_{\boldsymbol{w}}(-2\boldsymbol{w}^\mathsf{T}\boldsymbol{a}) = -2\boldsymbol{a} = -2\boldsymbol{X}^\mathsf{T}\boldsymbol{y}.$$

Finish the gradient

For the quadratic term, use

$$\boldsymbol{A} = \boldsymbol{X}^\mathsf{T}\boldsymbol{X}.$$

Since $\boldsymbol{A}$ is symmetric,

$$\nabla_{\boldsymbol{w}}(\boldsymbol{w}^\mathsf{T}\boldsymbol{X}^\mathsf{T}\boldsymbol{X}\boldsymbol{w}) = 2\boldsymbol{X}^\mathsf{T}\boldsymbol{X}\boldsymbol{w}.$$

Combining all terms:

$$\boxed{\nabla_{\boldsymbol{w}} J = -2\boldsymbol{X}^\mathsf{T}\boldsymbol{y} + 2\boldsymbol{X}^\mathsf{T}\boldsymbol{X}\boldsymbol{w}}.$$

Equivalently,

$$\nabla_{\boldsymbol{w}} J = -2\boldsymbol{X}^\mathsf{T}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}).$$

The least-squares gradient combines a linear term from the data and a quadratic term from the model.

Why $X^\top$ appears

The residual is in observation space:

$$\boldsymbol{r} = \boldsymbol{y} - \boldsymbol{X}\boldsymbol{w} \in \mathbb{R}^n.$$

But the gradient must be in parameter space:

$$\nabla_{\boldsymbol{w}} J \in \mathbb{R}^d.$$

The transpose has the right shape:

$$\boldsymbol{X}^\top \in \mathbb{R}^{d \times n}.$$

Therefore,

$$\boldsymbol{X}^\top \boldsymbol{r} \in \mathbb{R}^d.$$

This maps residual information back to the coordinates of $\boldsymbol{w}$. The gradient can be written as

$$\nabla_{\boldsymbol{w}} J = -2\boldsymbol{X}^\top \boldsymbol{r}.$$

Dimension checking explains why the transpose is needed.

Normal equations

At an unconstrained optimum,

$$\nabla_{\boldsymbol{w}} J = \mathbf{0}.$$

Therefore,

$$-2\boldsymbol{X}^{\mathsf{T}}\boldsymbol{y} + 2\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X}\hat{\boldsymbol{w}} = \mathbf{0}.$$

Divide by 2:

$$-\boldsymbol{X}^{\mathsf{T}}\boldsymbol{y} + \boldsymbol{X}^{\mathsf{T}}\boldsymbol{X}\hat{\boldsymbol{w}} = \mathbf{0}.$$

Rearranging gives

$$\boxed{\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X}\hat{\boldsymbol{w}} = \boldsymbol{X}^{\mathsf{T}}\boldsymbol{y}}.$$

These are the normal equations.

Ridge gradient

Ridge adds a penalty:

$$J_{\text{ridge}}(\boldsymbol{w}) = \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}\|_2^2 + \lambda \boldsymbol{w}^\mathsf{T}\boldsymbol{w}.$$

We already know

$$\nabla_{\boldsymbol{w}} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}\|_2^2 = -2\boldsymbol{X}^\mathsf{T}\boldsymbol{y} + 2\boldsymbol{X}^\mathsf{T}\boldsymbol{X}\boldsymbol{w}.$$

The penalty contributes

$$\nabla_{\boldsymbol{w}} (\lambda \boldsymbol{w}^\mathsf{T}\boldsymbol{w}) = 2\lambda \boldsymbol{w}.$$

Therefore,

$$\nabla_{\boldsymbol{w}} J_{\text{ridge}} = -2\boldsymbol{X}^\mathsf{T}\boldsymbol{y} + 2\boldsymbol{X}^\mathsf{T}\boldsymbol{X}\boldsymbol{w} + 2\lambda \boldsymbol{w}.$$

Ridge equations

Setting the ridge gradient to zero gives

$$-2\boldsymbol{X}^\top \boldsymbol{y} + 2\boldsymbol{X}^\top \boldsymbol{X}\hat{\boldsymbol{w}} + 2\lambda \hat{\boldsymbol{w}} = \boldsymbol{0}.$$

Divide by 2 and collect terms containing $\hat{\boldsymbol{w}}$:

$$\boldsymbol{X}^\top \boldsymbol{X}\hat{\boldsymbol{w}} + \lambda \boldsymbol{I}\hat{\boldsymbol{w}} = \boldsymbol{X}^\top \boldsymbol{y}.$$

Therefore,

$$\boxed{(\boldsymbol{X}^\top \boldsymbol{X} + \lambda \boldsymbol{I})\hat{\boldsymbol{w}} = \boldsymbol{X}^\top \boldsymbol{y}}.$$

If the penalty is centered at $\boldsymbol{w}_0$, the penalty gradient becomes

$$2\lambda(\boldsymbol{w} - \boldsymbol{w}_0).$$

Ridge uses the same derivative rules as least squares, with one additional penalty gradient.